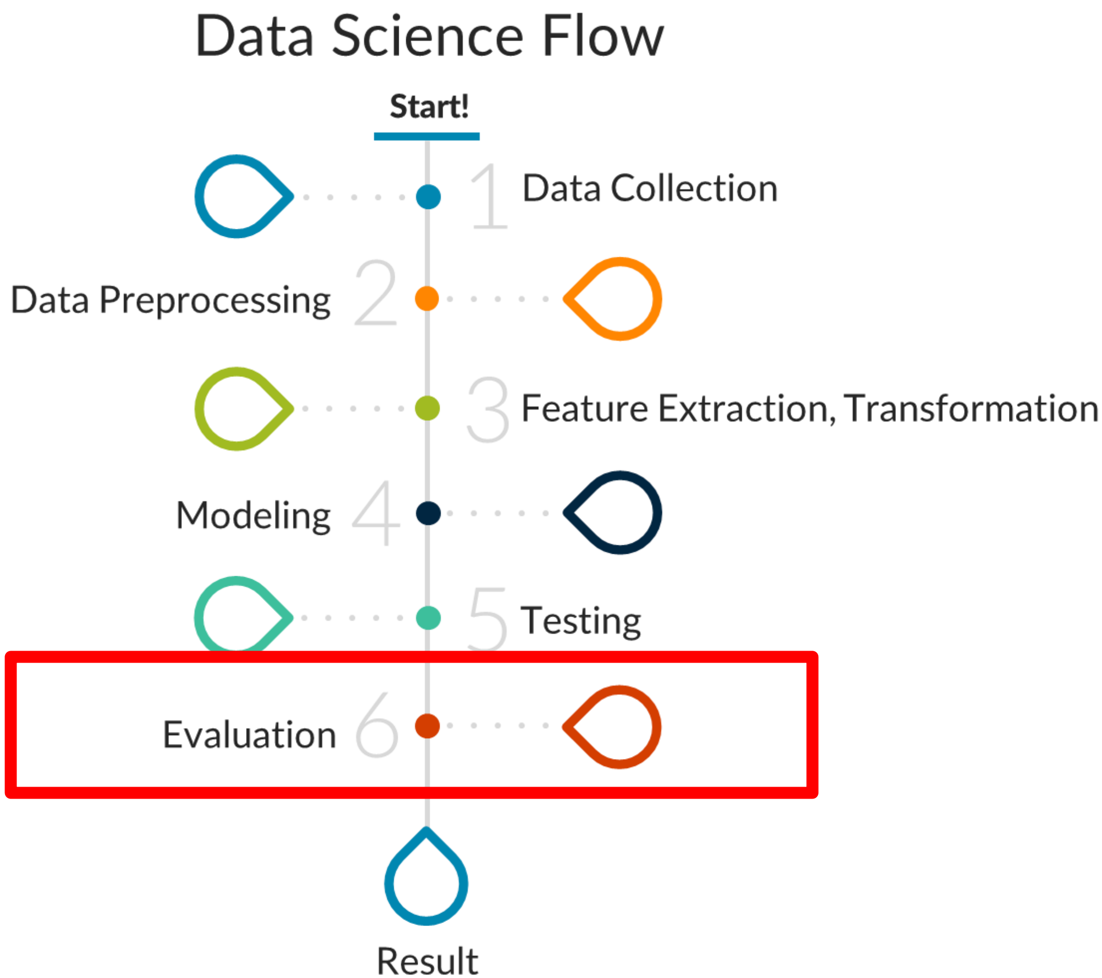
Metrics

Outline

- Introduction
- Classification
- Clustering
- Segmentation
- Object Detection

Introduction

Introduction



Introduction

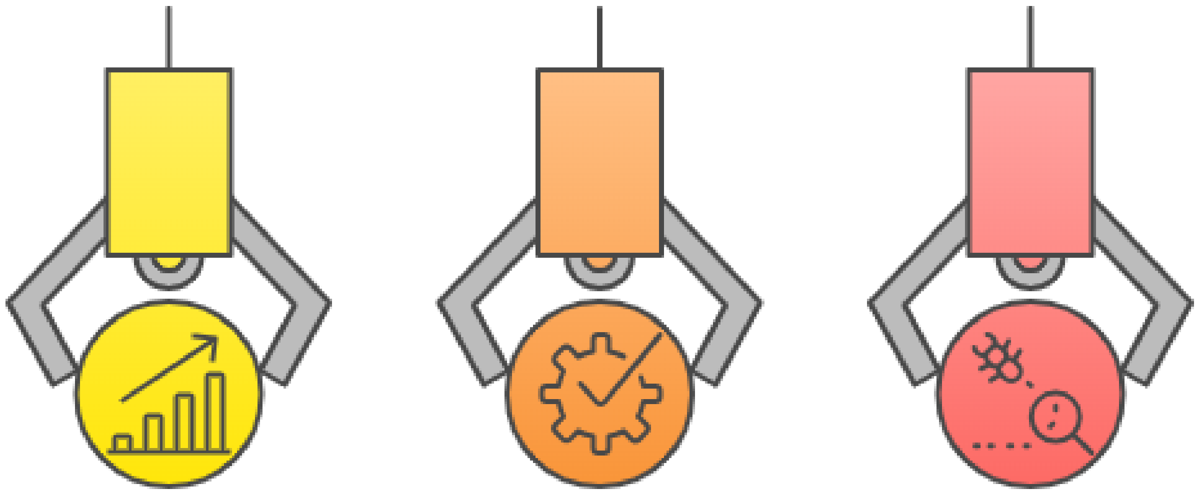
- Model evaluation: quantifying the quality of predictions
- **The Role of Metrics**: Metrics serve as the core of the model training loop, guiding the direction of iterative improvements.

"If you can't measure it, you can't improve it."

- Peter Drucker

Introduction

Importance of Metrics



Quantifying Performance

Metrics convert abstract model quality into comparable figures to guide development.

Aligning Objectives

Metrics ensure that optimization aligns with specific business logic, such as cost reduction or minimizing false alarms.

Detecting Flaws

Various metrics help identify failures within specific classes or edge cases.

Introduction

Case Study: Rare Disease Screening

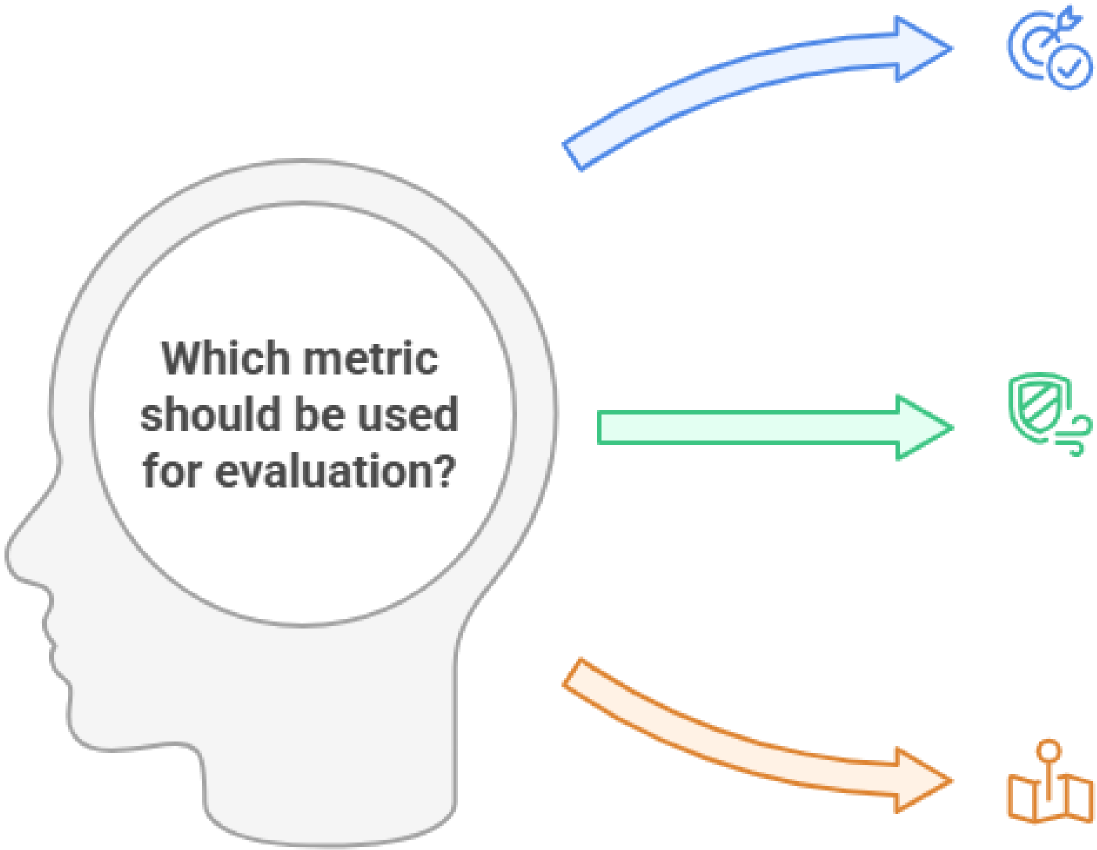
10,000	**100**	**9,900**
Total Samples	True Positive (1%)	True Negative (99%)

👉 **If the model predicts "Healthy" for everyone:**

- It misses every single sick patient...
- Yet, Accuracy is still **99%**

Conclusion: Accuracy can be misleading in imbalanced datasets.

Introduction



Correctness

Ensures predictions align with facts, suitable for general accuracy assessment. (e.g., Accuracy, Error Rate)

Robustness

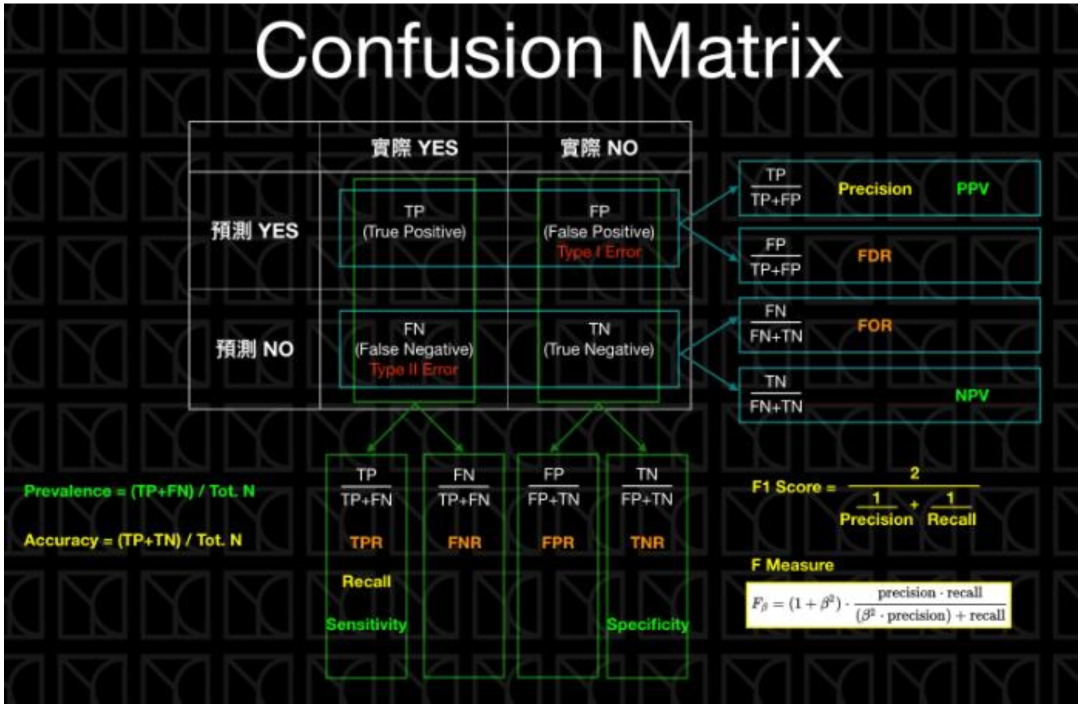
Maintains accuracy with extreme class proportions, ideal for imbalanced datasets. (e.g., Dice, F1-Score, mAP)

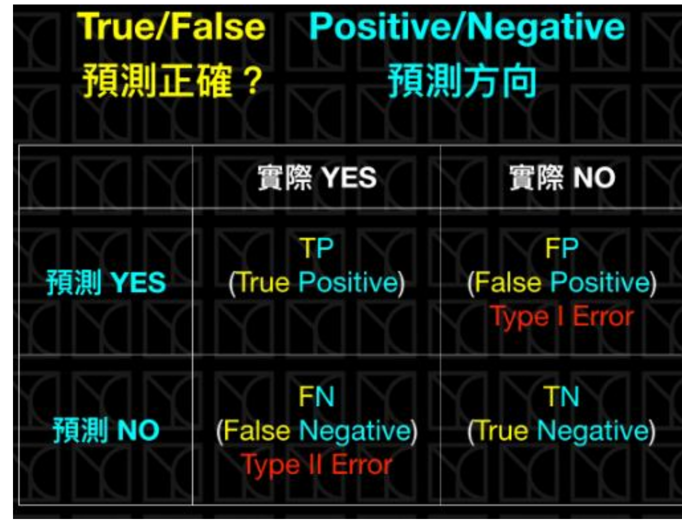
Localization

Precisely identifies position and shape, best for object detection and segmentation. (e.g., IoU, Boundary Dice)

Metrics of Classification Tasks

Confusion Matrix





- True and False represents the prediction result is correct or not.
- Positive and Negative represents the prediction is positive or negative.

https://www.ycc.idv.tw/confusion-matrix.html

Confusion Matrix

		ACTUAL	
		Negative	Positive
PREDICTION	Negative	60	8
	Positive	22	10

- **True Positive (TP)** — model correctly predicts the positive class (prediction and actual both are positive). In the above example, **10 people** who have tumors are predicted positively by the model.
- **True Negative (TN)** — model correctly predicts the negative class (prediction and actual both are negative). In the above example, **60 people** who don't have tumors are predicted negatively by the model.
- **False Positive (FP)** — model gives the wrong prediction of the negative class (predicted-positive, actual-negative). In the above example, **22 people** are predicted as positive of having a tumor, although they don't have a tumor. FP is also called a **TYPE I** error.
- **False Negative (FN)** — model wrongly predicts the positive class (predicted-negative, actual-positive). In the above example, **8 people** who have tumors are predicted as negative. FN is also called a **TYPE II** error.

Example

- Unlock your iphone by the fingerprint.

	Ground Truth Yes	Ground Truth No
Prediction Yes	a	d
Prediction No	c	b

(a) Your iphone unlock your fingerprint.
(b) Your iphone cannot unlock your friend's fingerprint.
(c) Your iphone cannot unlock your fingerprint.
(d) Your iphone unlock your friend's fingerprint.

Confusion Matrix

- Accuracy= (TP+TN)/(TP+FP+FN+TN)
  - Defined as the ratio of correctly predicted examples by the total examples.
  - If the ground truth yes is few, the accuracy is not accurate. For example, prediction of credit card fraud can easily reach more than 99%.
- Recall(召回率) = TP/(TP+FN)
  - Out of the total positive, what percentage are predicted positive. It is the same as TPR (true positive rate).
  - Advertisement
- Precision(精確率) = TP/(TP+FP)
  - Out of all the positive predicted, what percentage is truly positive.
  - Access control system

True/False 預測正確？ Positive/Negative 預測方向

	實際 YES	實際 NO
預測 YES	TP (True Positive)	FP (False Positive) Type I Error
預測 NO	FN (False Negative) Type II Error	TN (True Negative)

Confusion Matrix

- F Measure

F Measure

$$F_\beta = (1+\beta^2) \cdot \frac{\text{precision} \cdot \text{recall}}{(\beta^2 \cdot \text{precision}) + \text{recall}}$$

- β=1 => F1-score
  - If Precision and Recall are important, we can use "F1-Score."
  - F1-score = 2 * Precision * Recall / (Precision + Recall)
- If precision is more important， β is smaller.
- If recall is more important, β is bigger.
- If β=0, it is precision.
- If β is infinity, it is recall.

Average

- For binary classification, we can focus on positive class, such as junk mail, cancer prediction.
- For multi-class classification, we can use the macro, micro and weighted average of recall, precision, f1 score.
- These evaluation metrics can help you choose the best model.

Average

- average=micro
  - Compute f1 by considering total true positives, false negatives and false positives (no matter of the prediction for each label in the dataset)
- average=macro
  - Compute f1 for each label, and returns the average without considering the proportion for each label in the dataset.
- average=weighted
  - Compute f1 for each label, and returns the average considering the proportion for each label in the dataset.

Example

	Predicted (what our model says))				
Actual (what the data says)	CLASSES	A	B	C	Row totals
	A	5	2	3	10
	B	2	6	0	8
	C	3	2	2	7
	Column Totals	10	10	5	25

Diagonal numbers are rightly classified observations

Total number of observations/ records

#MLmuse

CLAIRVOYANT

Accuracy

- Accuracy of the model

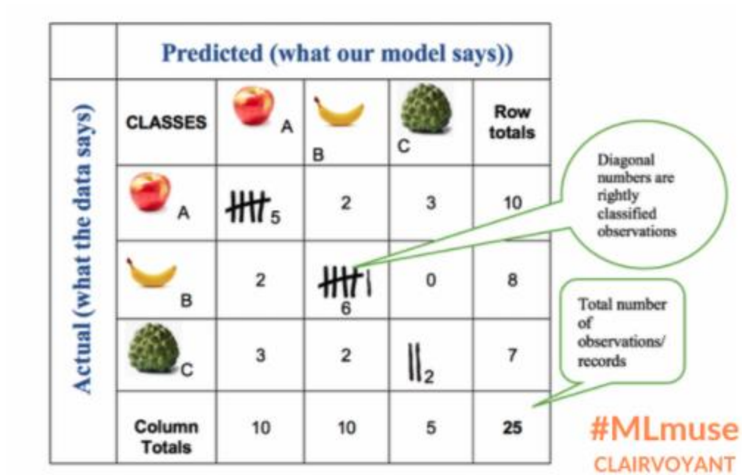
= No. of correct predictions by the model/total observations

= Sum (diagonal values of confusion matrix)/ total observations

= (5+6+2)/25

=13/25

=0.25



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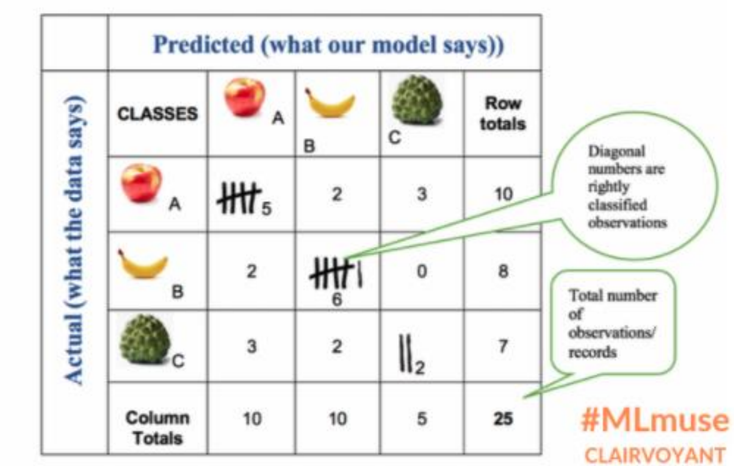
Recall

- True Positive Rate or Sensitivity
- Recall for class B :From all the actual instances of class B (banana), how often it correctly predicts as B (banana)
- Recall for class B = out of the total Bs (bananas = 8) in the data, how many Bs did the model identify correctly (6)

= 6/8 = 0.75



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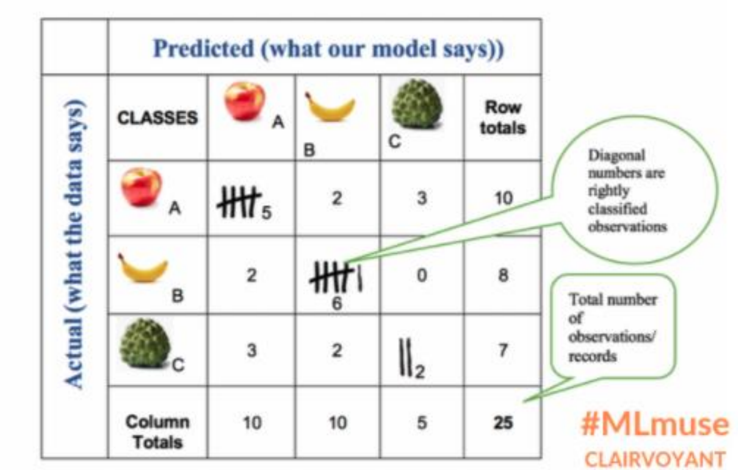
Precision

- Precision for class B :how often is the model correct when it predicts as B?
- Precision for B: out of the total observations predicted as Bs (10) by the model, how many are correct Bs (6).

= 6/10 = 0.6



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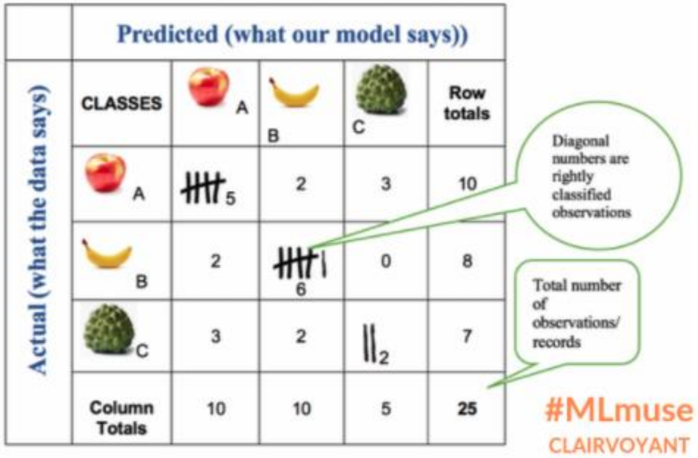
F1-Score

- F1 score of class B will be (2 x Recall x precision)/(Recall + Precision) =0.75*0.6*2/(0.75+0.6) = 0.67



Mirco-Average Precision

- **Micro-average precision** is a metric used to evaluate the precision of a multi-class or multi-label classification model by treating all classes equally, aggregating the total number of true positives and false positives across all classes before calculating the precision.

$$\text{Micro-average precision} = \frac{\sum_c \text{True Positives}_c}{\sum_c \text{True Positives}_c + \sum_c \text{False Positives}_c}$$

Mirco-Average Precision

- **When to Use Micro-average Precision**
  - Class distributions are highly imbalanced.
  - The performance on each instance is more important than performance on each individual class.

Mirco-Average Recall

- **Micro-average recall** is a metric used in multi-class or multi-label classification to measure the recall across all classes by treating each instance equally, regardless of class.
  - Instead of calculating recall separately for each class and averaging them, micro-average recall aggregates true positives and false negatives across all classes and then calculates the recall.

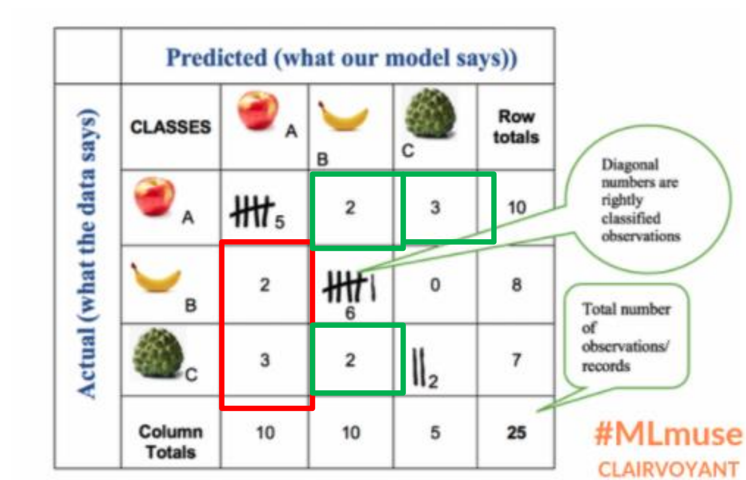
$$\text{Micro-average recall} = \frac{\sum_c \text{True Positives}_c}{\sum_c \text{True Positives}_c + \sum_c \text{False Negatives}_c}$$

Mirco-Average Recall

- **When to Use Micro-average Recall**
- You care about the overall performance across all instances, not specific class-level performance.
- Class distributions are imbalanced, and you want to avoid biased results due to small or large classes.
- In contrast, **macro-average recall** would calculate recall for each class separately and then average these values, which could overemphasize the performance on classes with fewer instances.

Mirco-Average Recall

- In the Micro-average method, you sum up the individual class’s true positives, false positives, and false negatives of the system for different sets and then apply them to get the score.
- (5+6+2)/((5+6+2)+(5+4+3)) =0.52



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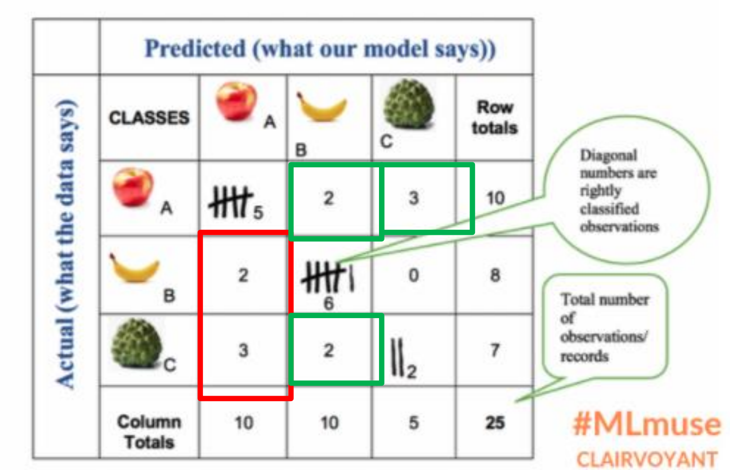
Marco-Average Recall

- It is the simple mean of scores of all classes.
- So, macro- average recall is the mean of the recalls of classes A, B and C.
- (0.5+0.75+0.29)/3 =0.513



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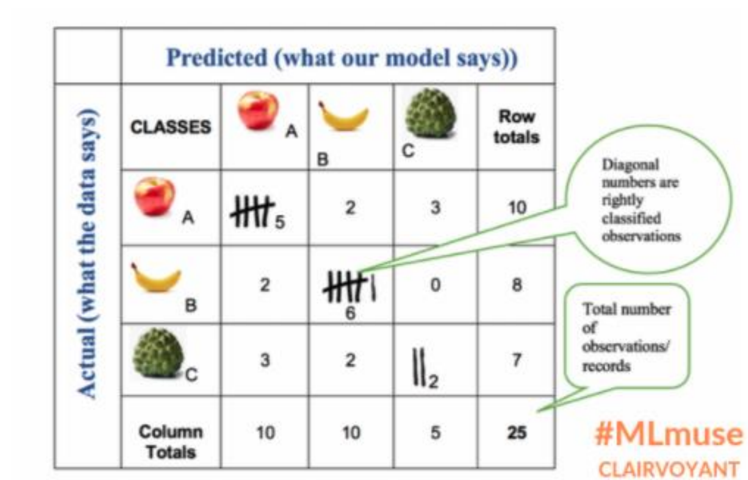
Weighted-Average Recall

- The sum of the scores of all classes after multiplying their respective class proportions.
- (10*0.5+8*0.75+7*0.29)/25 = 0.5212



Metrics of Clustering Tasks

Distance Measure Method

- **Euclidean distance measure**:
  - Simplest
  - The Euclidean distance between point *p* and *q* in N-dimensional space is given as:

$$d = (p, q) = \sqrt{\sum_{i=1}^{N} (p_i - q_i)^2}$$

- **Cosine distance measure:**
  - Finds the cosine of angle between two vectors (vectors drawn from origin to the points.)

$$d = \cos(\theta) = \frac{A \cdot B}{||A||\,||B||}$$

Distance Measure Method

- **Manhattan distance measure:**
  - The sum of the absolute differences of the coordinates of two points.

➢ If $P(x_1, y_1)$and $Q(x_2, y_2)$,

Manhattan Distance=| $x_2 - x_1$ |+| $y_2 - y_1$ |

➢ For n-dimensional space：coordinates $P(p_1, p_2, \dots, p_n)$, $Q(y_1, y_2, \dots, y_n)$

$$d = (p, q) = \sum_{\mathrm{i}=1}^{\mathrm{N}} |\mathrm{p_i} - \mathrm{q_i}|$$

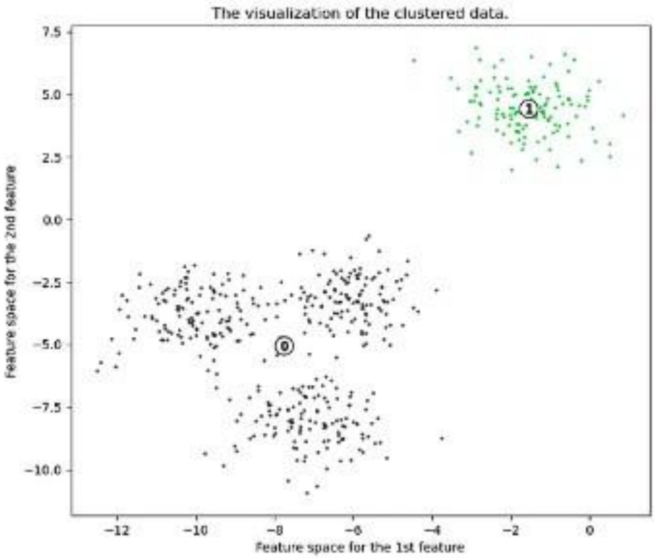
Evaluation Metric of Clustering

- Clustering performance: Silhouette Score
  - The Silhouette Score and Silhouette Plot are used to measure the separation distance between clusters.
  - The range is [-1, 1].
    - A score close to **1** indicates that the object is well-clustered (far from neighboring clusters).
    - A score near **0** means the object is on or near the boundary between clusters.
    - A negative score suggests that the object may have been assigned to the wrong cluster.
- A higher Silhouette Score (**S**) indicates clearer separation between clusters.

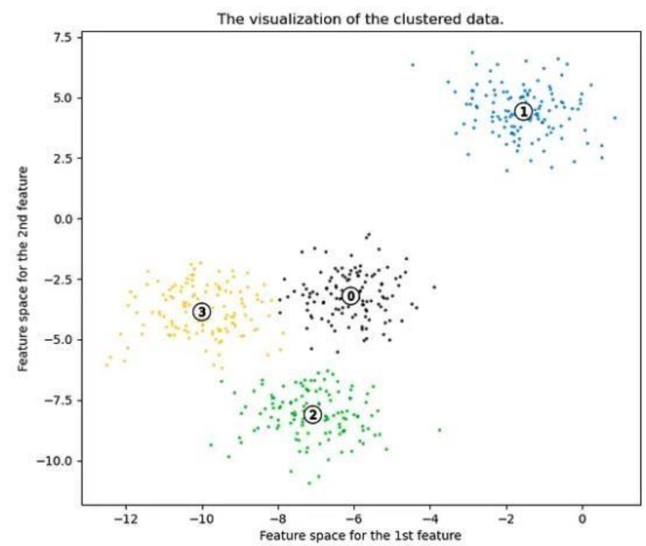
$$S = \frac{b - a}{max(a, b))}$$

- **a** is the average distance between points in the same cluster.
- **b** is the average distance between points in different clusters.

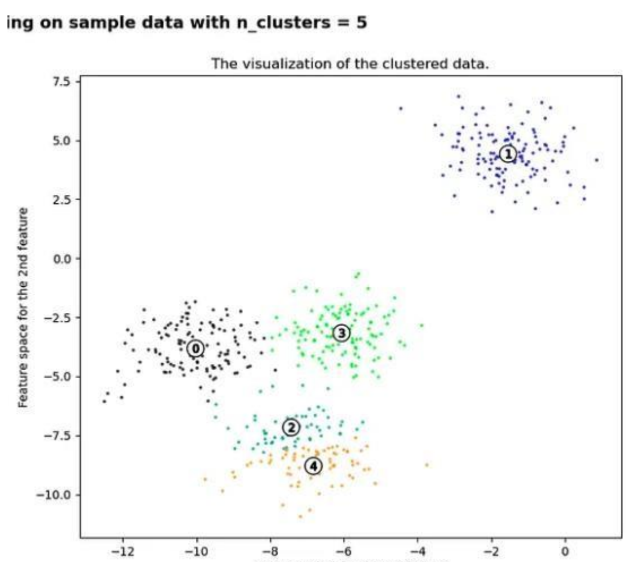
0.7



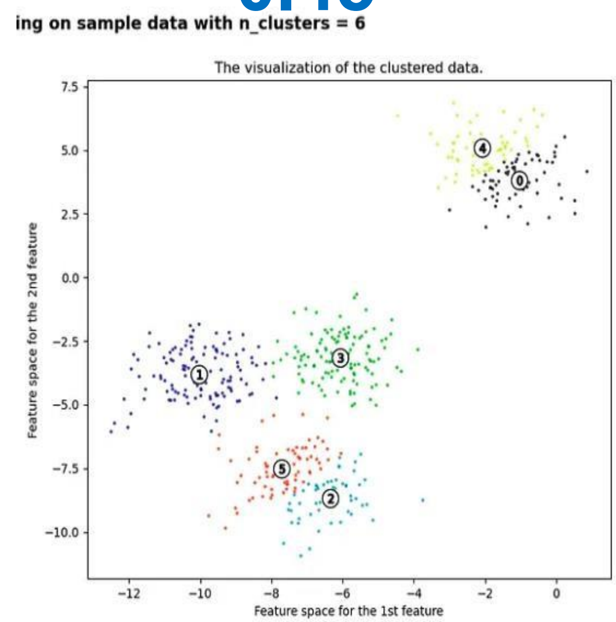
0.65



0.56



0.45

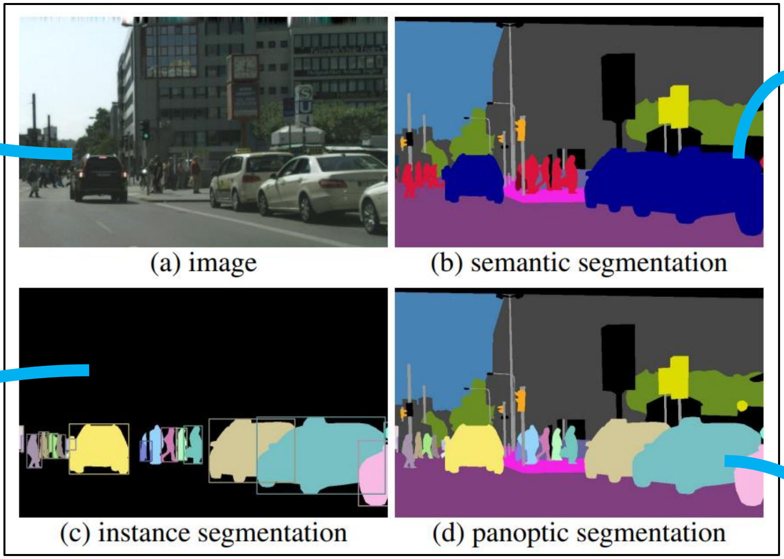


Metrics of Segmentation Tasks

Segmentation Types

Ground Truth

Semantic Segmentation (per-pixel class labels)



Instance Segmentation (per-object mask and class label)

Panoptic Segmentation (per-pixel class+ instance labels)

Kirillov, A., He, K., Girshick, R., Rother, C., & Dollár, P. (2019). Panoptic segmentation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 9404-9413).

Pixel-based Metrics

- Pixel Accuracy(PA)
  - Calculates the ratio of correctly classified pixels to the total pixels across the image with *k* classes.

$$PA = \frac{\sum_{i=1}^{k} n_{ii}}{\sum_{i=1}^{k} gt_i}$$

n_{ii} stands for Number of pixels of class i correctly predicted as class i
gt_i stands for Total number of pixels belonging to class i

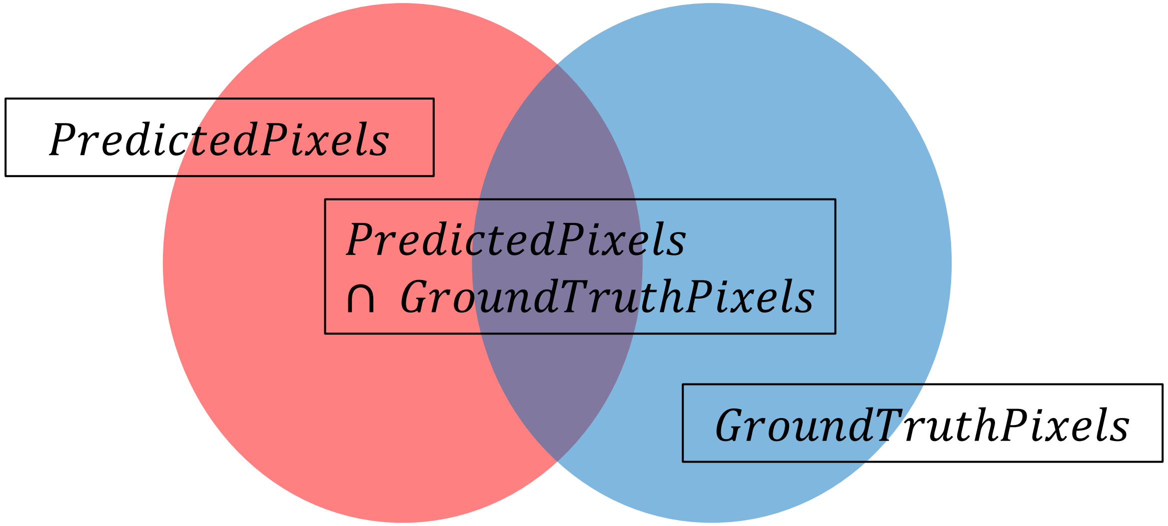
- Mean Accuracy(MA)
  - Calculates the average of the accuracies of all individual classes.

$$MA = \frac{1}{k}\sum_{i=1}^{k} \frac{n_{ii}}{gt_i}$$

Region-based Metrics

- Dice Coefficient
  - Standardizing accuracy for complex shapes and small targets.
  - Dice measures spatial overlap between the Predicted Mask and the Ground Truth.

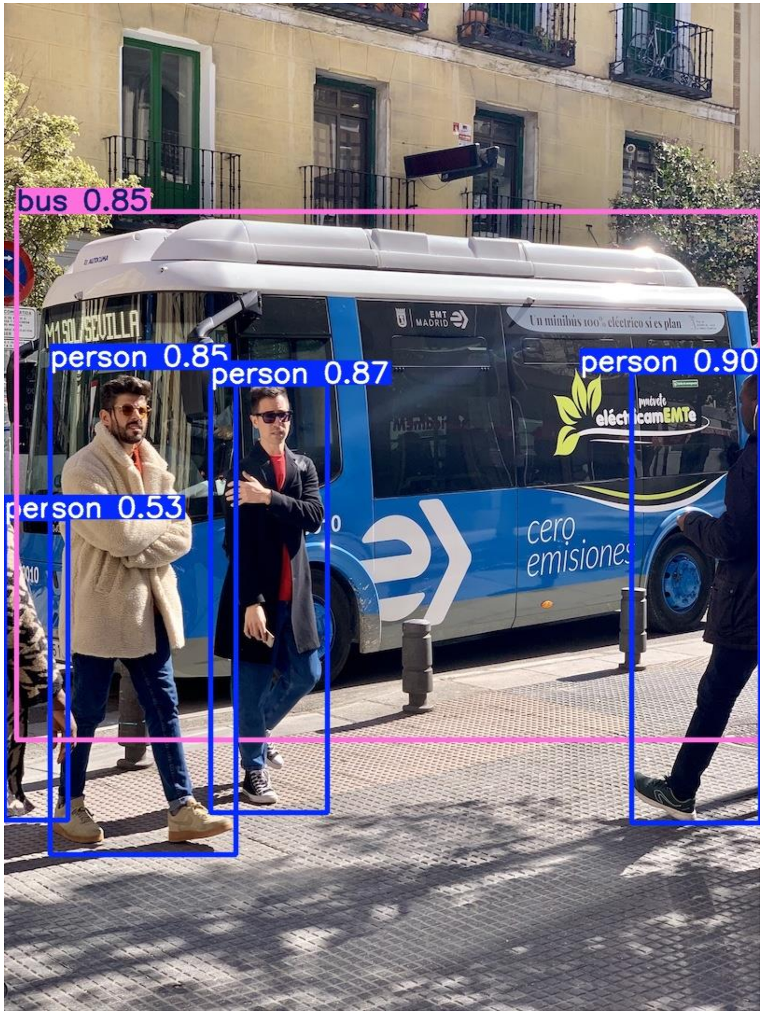
$$Dice = \frac{2|PredictedPixels \cap GroundTruthPixels|}{|PredictedPixels| + |GroundTruthPixels|}$$



Metrics of Object Detection Tasks

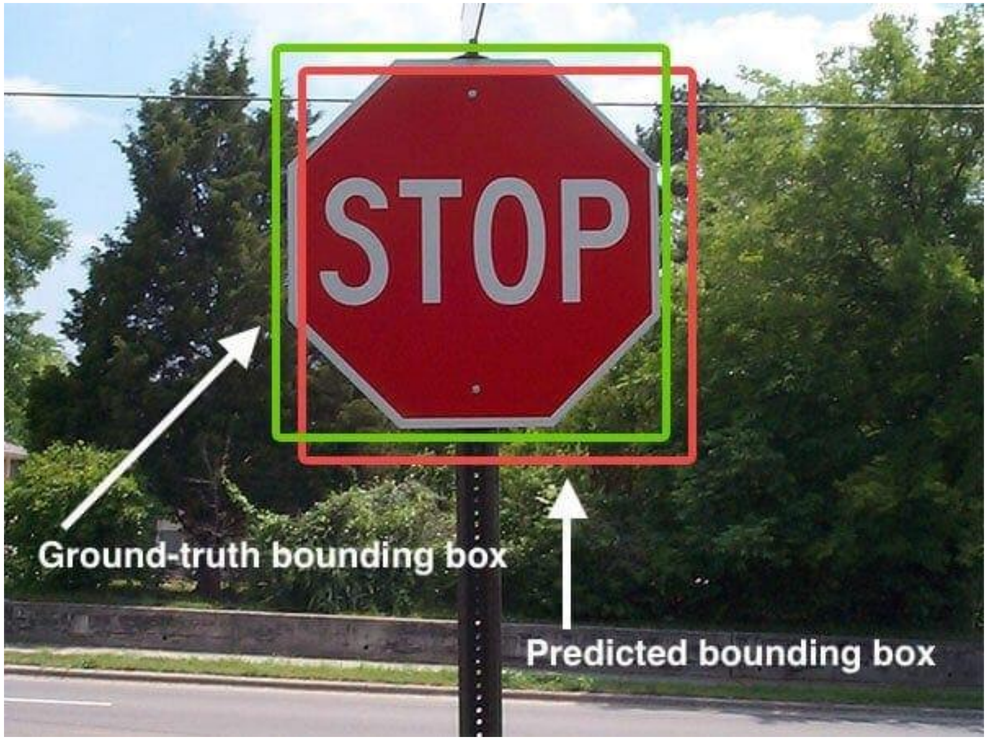
Object Detection Tasks

- **Classification + Localization**
- Unlike image classification which identifies a single global label.
- Object Detection answers two questions simultaneously:
- **What:** Identifying the class of the object (e.g., Dog, Car, Person).
- **Where:** Precisely locating the object with a Bounding Box.



Intersection over Union (IoU)

- Intersection over Union (IoU) is used to evaluate the performance of object detection by comparing the ground truth bounding box to the predicted bounding box.



Intersection over Union (IoU)

$$\text{IoU} = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$

An IoU score > *0.5* is normally considered a "good" prediction.

mean Average Precision (mAP)

- mAP provides a single comprehensive score that balances classification accuracy and localization precision across all object categories.

mAP example

- The whole dataset contains 5 apples only.
- We collect all the predictions made for apples in all the images and rank it in descending order according to the predicted confidence(or probability) level.
- The second column indicates whether the prediction is correct or not. In this example, the prediction is correct if IoU ≥ 0.5.

Rank	Correct?	Precision	Recall
1	True	1.0	0.2
2	True	1.0	0.4
3	False	0.67	0.4
4	False	0.5	0.4
5	False	0.4	0.4
6	True	0.5	0.6
7	True	0.57	0.8
8	False	0.5	0.8
9	False	0.44	0.8
10	True	0.5	1.0

mAP example

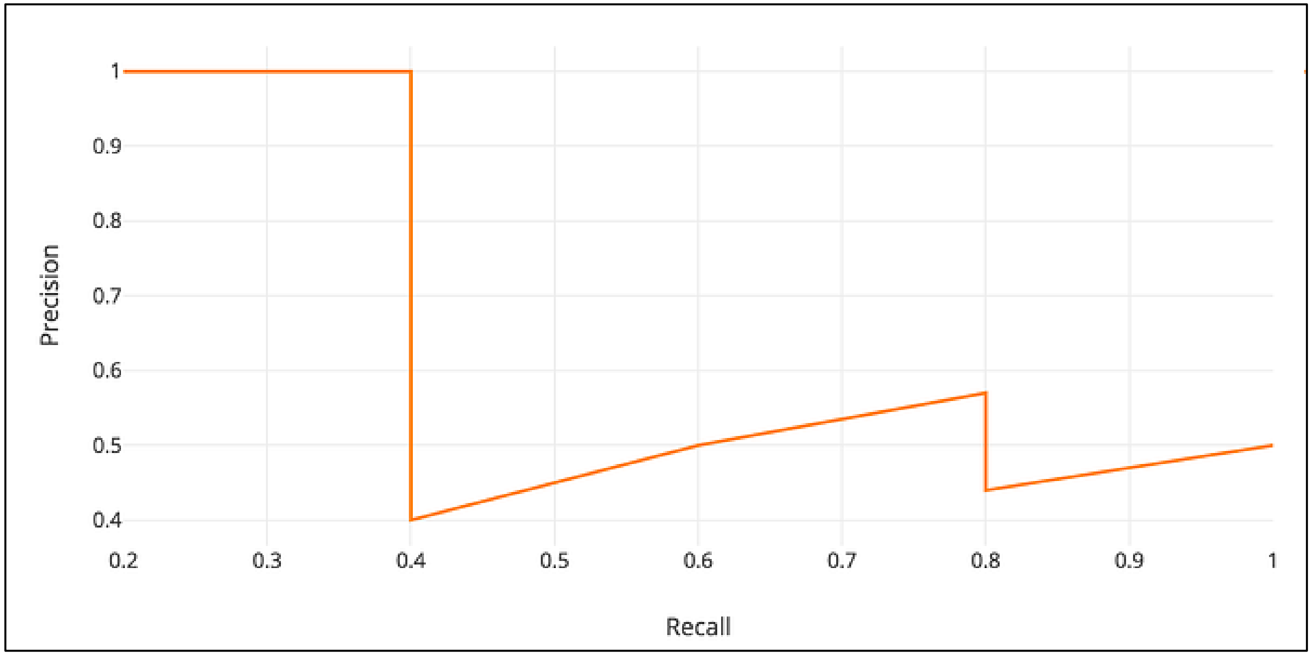
Rank	Correct?	Precision	Recall
1	True	1.0	0.2
2	True	1.0	0.4
3	False	0.67	0.4
4	False	[redacted]	[redacted]
5	False	[redacted]	[redacted]
6	True	[redacted]	[redacted]
7	True	[redacted]	[redacted]
8	False	[redacted]	[redacted]
9	False	[redacted]	[redacted]
10	True	[redacted]	[redacted]

- $Precision = \frac{\mathrm{TP}}{\mathrm{TP+FP}} = \frac{2}{2+1} = \frac{2}{3} = 0.67$
- $\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP+FN}} = \frac{2}{5} = 0.4$

TRY IT!

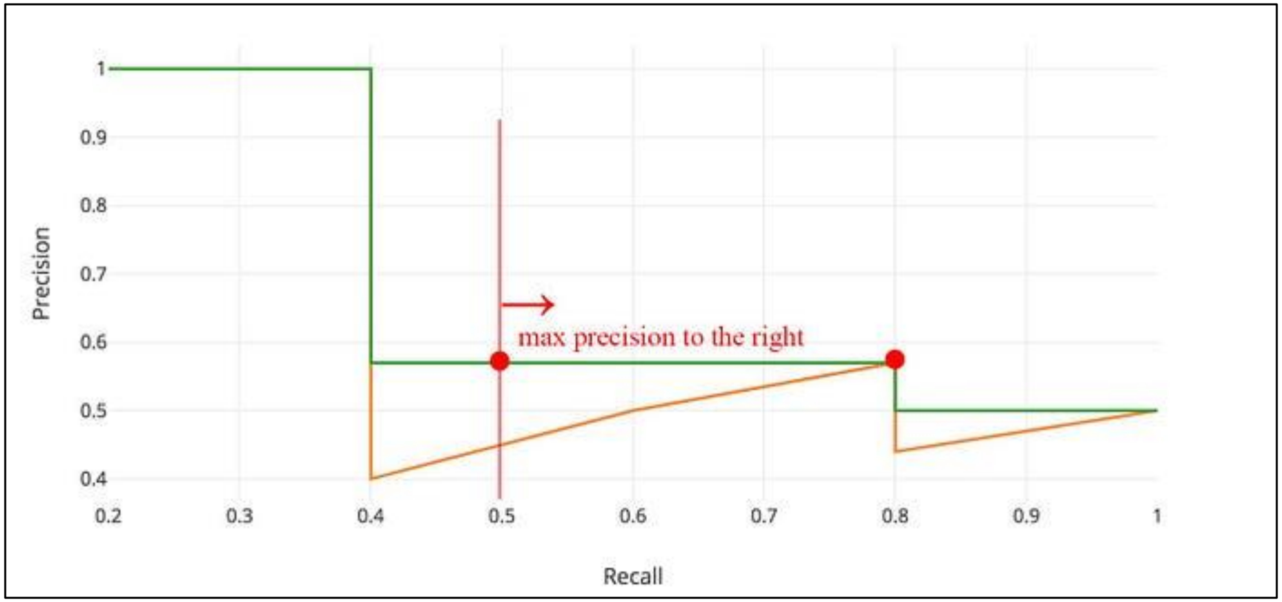
mAP example

- plot the precision against the recall value to see this zig-zag pattern.



mAP example

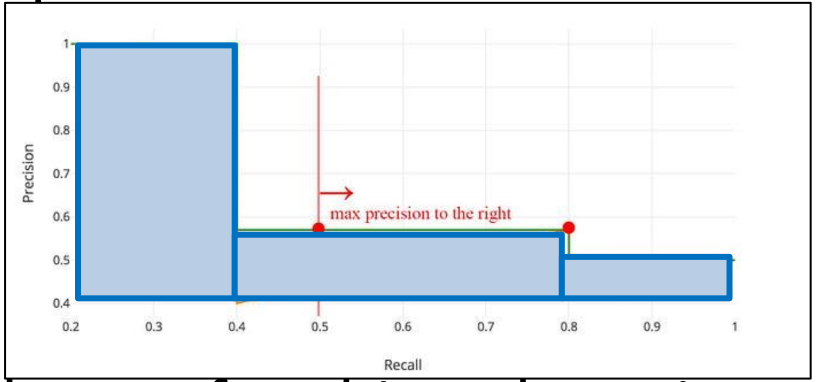
- Before calculating AP for the object detection, we often smooth out the zigzag pattern first.



- Sorting simulates the trade-off between precision and recall, while smoothing ensures a robust, monotonic curve that accurately reflects the model's true performance.

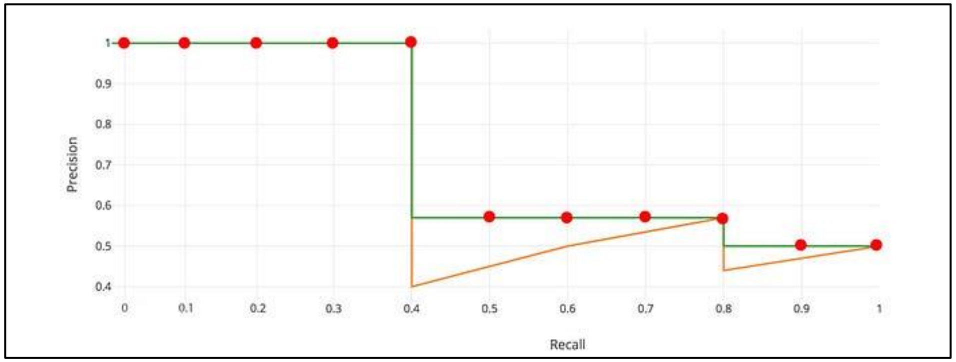
mAP example

- The general definition for the Average Precision (AP) is finding the area under the precision-recall curve above.
- $AP = \int_0^1 p(r)dr$



- PASCAL VOC is a popular dataset for object detection. In Pascal VOC 2008, an average for the 11-point interpolated AP is calculated.

$$AP = \frac{1}{11} \sum_{r \in \{0.0, 0.1, \ldots, 1.0\}} AP_r$$
$$= \frac{1}{11} \sum_{r \in \{0.0, 0.1, \ldots, 1.0\}} P_{interp}(r)$$



mAP example

- mAP is the mean of APs across all ***i*** classes in the dataset.

$$mAP = \frac{1}{N}\sum_{i=1}^{N} AP_i$$

- **mAP@0.5** corresponds to the average AP for IoU at 0.5.
- **mAP@[.5:.95]** corresponds to the average AP for IoU from 0.5 to 0.95 with a step size of 0.05.

Metrics Lab Activity

Scenario

- Today, we will challenge ourselves with the real-world scenario -- Credit Card Fraud Detection dataset.

1. Experimental Environment Setup

- Run the following Python code to load the dataset and train a baseline model. We are using a Random Forest classifier to detect these rare fraudulent events.
- Run the cell
-
```python
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn import metrics
```

- Complete the code and run the cell
-
```python
# TO DO: Loading a real-world Credit Card Fraud dataset
# hint:data = pd.read_csv(....)

```

1. Experimental Environment Setup

- Run the cell and check your output

```python
# Observing the distribution: 0 = Normal, 1 = Fraud
print(f"Data Distribution:\n{data['Class'].value_counts(normalize=True)}")
```

```
Data Distribution:
Class
0    0.998273
1    0.001727
Name: proportion, dtype: float64
```

```python
# Preprocessing and Splitting
X = data.drop('Class', axis=1)
y = data['Class']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, random_state=42)

# Training a Random Forest Model
clf = RandomForestClassifier(n_estimators=100, random_state=42)
clf.fit(X_train, y_train)
y_pred = clf.predict(X_test)
```

2.Implementation & Evaluation

- Complete the code and run the cell

```python
#TO DO: Complete the code and run the cell
def evaluate_model(y_true, y_prediction):
    # hint: Call the appropriate sklearn functions ---
    acc =     # Function for Accuracy
    prec =   # Function for Precision
    rec =  # Function for Recall
    f1 =  # Function for F1-score


    print(f"1. Accuracy:  {acc:.4f}")
    print(f"2. Precision: {prec:.4f}")
    print(f"3. Recall:    {rec:.4f}")
    print(f"4. F1-Score:  {f1:.4f}")

    print("\nConfusion Matrix:")
    print(metrics.confusion_matrix(y_true, y_prediction))

evaluate_model(y_test, y_pred)
```

```
1. Accuracy:  [redacted]
2. Precision: [redacted]
3. Recall:    [redacted]
4. F1-Score:  [redacted]

Confusion Matrix:
[[[redacted]  [redacted]]
 [ [redacted]  [redacted]]]
```

3. Worksheet

- **Q1: Observations on Accuracy**

  In credit card fraud, a model predicting "Healthy/Normal" for everyone can easily reach >99% Accuracy. Given your results, why is Accuracy considered a "misleading" metric in this specific case?

- **Q2: Precision vs. Recall Trade-off**

  -Recall (Sensitivity): How many actual frauds did we catch?

  -Precision: How many of our fraud alerts were actually correct?

  In a banking environment, which error is more "expensive": A False Positive (Type I Error) or a False Negative (Type II Error) and why?

- **Q3: The Final Decision Metric**

  If you need to balance both Precision and Recall equally, which composite metric should you use? If catching every single fraud is the top priority (even with some false alarms), how should you adjust the *ß* value in the F-Measure?

Submission

- Please create a report to briefly explain the results and questions in worksheet. Attach the code that you wrote and the screenshots of the outputs.
- Submit the report **before the specified time.**